Occupational Medicine: The Importance of the History & Physical Examination

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OD3

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Objectives: The Student will be able to:

Understand	Understand the historic compromise and the “No Fault” Rule
Comprehend	Comprehend the Evolution of Workers’ Compensation in California
Take	Take a complete occupational history
Define	Define the difference between an aggravation and an exacerbation
Understand and apply	Understand and apply the legal definition of causation
Apply	Apply the concept of presumptions
Explain	Explain how psychiatric injuries are defined
Identify	Identify the different types of Occupational Physicals
Apply	Apply evidence-based medicine to decisions on occupational physicals

Triangle Shirt Waist Company Fire

141 MEN AND GIRLS DIE IN WAIST FACTORY FIRE; TRAPPED HIGH UP IN WASHINGTON PLACE BUILDING; STREET STREWN WITH BODIES; PILES OF DEAD INSIDE

The Flames Spread with Deadly Rapidity Through Flimsy Material Used in the Factory.

600 GIRLS ARE HEMMED IN

When Elevators Stop Many Jump to Certain Death and Others Perish in Fire-Filled Lofts.

STUDENTS RESCUE SOME

Help Them to Roof of New York University Building, Keeping the Panic-Stricken in Check.

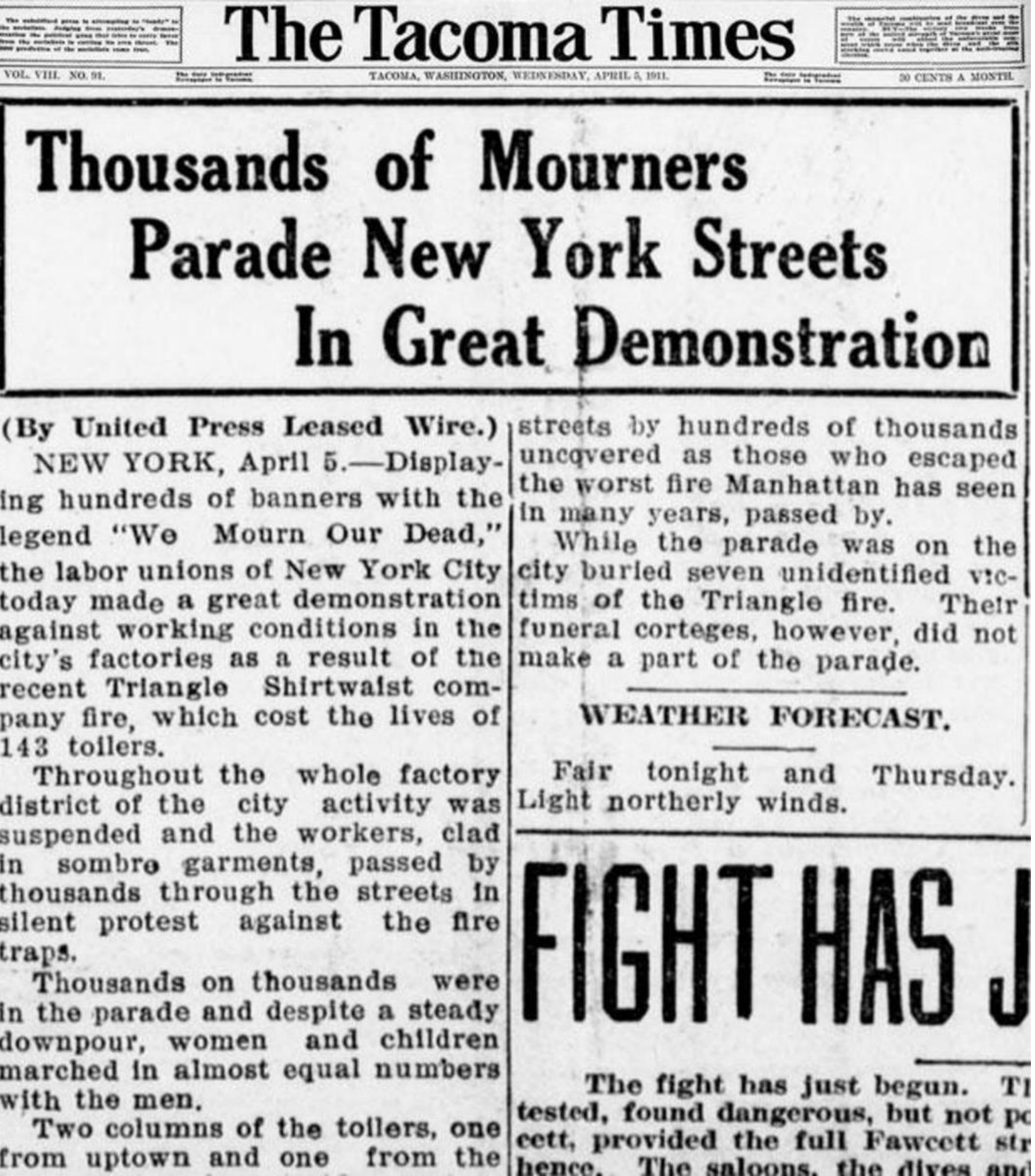
ONE MAN TAKEN OUT ALIVE

Plunged to Bottom of Elevator Shaft and Lived There Amid Flames for Four Hours.

ONLY ONE FIRE ESCAPE

Coroner Declares Building Laws

The Burning Building at 23 Washington Place.



The Tacoma Times

VOL. VIII. NO. 91. TACOMA, WASHINGTON, WEDNESDAY, APRIL 5, 1911. 30 CENTS A MONTH.

Thousands of Mourners Parade New York Streets In Great Demonstration

(By United Press Leased Wire.)

NEW YORK, April 5.—Displaying hundreds of banners with the legend "We Mourn Our Dead," the labor unions of New York City today made a great demonstration against working conditions in the city's factories as a result of the recent Triangle Shirtwaist company fire, which cost the lives of 143 toilers.

Throughout the whole factory district of the city activity was suspended and the workers, clad in sombre garments, passed by thousands through the streets in silent protest against the fire traps.

Thousands on thousands were in the parade and despite a steady downpour, women and children marched in almost equal numbers with the men.

Two columns of the toilers, one from uptown and one from the streets by hundreds of thousands uncovered as those who escaped the worst fire Manhattan has seen in many years, passed by.

While the parade was on the city buried seven unidentified victims of the Triangle fire. Their funeral corteges, however, did not make a part of the parade.

WEATHER FORECAST.

Fair tonight and Thursday. Light northerly winds.

FIGHT HAS J

The fight has just begun. Tl tested, found dangerous, but not pc cett, provided the full Fawcett str hence. The saloons, the dives and

The beginning of the Industrial Revolution

The Historical Compromise

Compromises by trading rights and benefits between employers and employees

Employee gives up right to pursue what might otherwise have been very large monetary award in exchange for a system that guarantees prompt delivery of benefits and provides legal protection against discrimination

Employer provides WC benefits regardless of fault, in exchange for protection against civil action by the employee

No Fault

- The employer is required to pay benefits no matter who caused the injury, as long as the injury **arose out of or occurred in the course of employment**
- **EXCEPTION:** Deliberately self-inflicted injuries are excluded from coverage; “serious and willful” misconduct of the injured employee, benefits are reduced by half; “serious and willful” misconduct by employer, compensation is increased by half

Exclusive Remedy

Unless the employer is uninsured, the worker cannot pursue other forms of recovery from the employer, even if the employer was grossly negligent

Assured & Fixed Benefits

- The WC System establishes defined benefits, which must be paid for by the employer.
- **NOTE:** WC awards are typically far less than comparable negligence awards in a civil suit, but a civil suit may take years to resolve.

Compensability

- WC System is a medically driven system
- Physician's opinion provides much of the basis for a claims administrator to start or deny benefits
- Guides key decision points in the system
- Requires physician to prepare detailed, complete, accurate, and unbiased reports
- Reports should be carefully written with conclusions that are consistent with the rest of the report

Complete Occupational Health History

- Information on all the jobs the worker has held
- Length of the worker’s employment
- Specific job duties
- How much time was spent on different tasks
- Any hazards (dusts, solvents, etc) to which the worker was exposed
- What kind of protective equipment was used

LABOR CODE Defines Injury as:

- Any injury or disease arising out of employment
- Any "derivative" injury caused by the treatment of an injury arising out of employment
- Any reaction to or side effect from preventative healthcare the employer provides to healthcare workers

Specific VS. Cumulative Trauma Injury

- An injury must either cause disability or result in a need for medical treatment
- A condition that does not cause lost work time or does not interfere with an employee’s ability to work is not considered an injury by the WC system
- A Specific Injury:
  - Occurs as the result of a single incident or exposure
  - A Cumulative Injury:
    - Results from repetitive trauma (mental or physical) over a period of time

Occupational Diseases

- A disease that in whole or in part was caused by work
- Can include diseases that, under other circumstances, may have occurred without a relationship to work
  - Example: a health care worker who contracts TB from a patient has sustained an occupational disease injury, even though the disease would not have been occupational if he had contracted the disease in a non-occupational setting

Occupational Diseases continued…

- Exposure to chemical agents, such as mercury or organic solvents
- Physical agents: noise, cold, radiation, or vibration
- Biological agents
- Repetitive motions such as kneeling, lifting, or typing

Excluded Injuries

- Injuries caused by the employee’s use of alcohol or illegal controlled substances
- Intentionally self-inflicted injuries
- Suicide
- Injuries resulting from altercations, in which the injured employee is the “initial physical aggressor”
- Injuries resulting from the employee’s commission of a felony, for which the employee has been convicted
- Injuries from off duty recreational activities
- Psychiatric injuries claimed after notice of termination/layoff

Compensable Injury

- An employer must provide compensation, without regard to negligence, for "any injury sustained by his or her employees arising out of and in the course of employment"
- There must be an "injury"
- There must be an employment relationship
- The injury must have been caused by the employment (AOE)
- The injury must occur in the course of employment (COE)

Aggravation

- Injury is considered when a worker who suffers an on-the -job aggravation of a non-industrial pre-existing disease or underlying condition
- An aggravation causes a temporary or permanent increase in disability that creates a new need for medical treatment or requires a change in the existing course of treatment
- “Flare up” or “recurrence” of a previous industrial injury or illness do not constitute a new injury

AOE

- The injured worker has the burden of proof to show by a preponderance of the evidence that an injury is work related
- The physician provides direct evidence on whether and how the activities of work led to the current injury
- Requires description of an incident and the resulting harm to the patient
- Requires detailed information about the how the injury occurred
- For cumulative trauma, the physician's medical opinion regarding the relationship between workplace risk factors and activities and the resulting disease or disability is critical

AOE

- There are many exposures, pathologies, and diseases for which the causal mechanism is not known
- The Lack of conclusive epidemiological or toxicological studies should not invalidate a worker's claim
- Physician's opinion based on combination of existing medical and scientific knowledge and the occupational and medical history of the individual worker to determine if there this is **"reasonable medical probability"** (more likely than not)

Causation

- **Direct Causation:** the work exposures are directly responsible for the health outcomes
- **Contributing Cause:** several factors led to the disease; work exposure is one of these factors
- **Acceleration:** the disease process is accelerated by virtue of work exposure. The date of onset of the disease is much sooner than it would have been in the absence of the exposure
- **Precipitation:** the work exposure "precipitates" the manifestation of the illness
- **Aggravation:** a medical condition may be present already, but work exposure makes it worse

City of Arroyo Grande v. WCAB (64 CCC 1147)

- During her normal workday, a school crossing guard suffers a heart attack and dies a week later. Medical records indicate that she had undiagnosed, non-industrial coronary artery disease at the time of her death. Her duties involved walking to the center of the street and holding a “Stop” sign as children crossed. She did this 10-15 times in 25-minute shifts. The reporting physicians concluded that, given her condition, even minimal activity could have caused the heart attack. The judge found industrial causation, which was upheld by the WCAB.

Causation

- Employer “takes employees as they find them”
- Employer cannot avoid liability for an occupational injury by claiming that the injury would not have happened if the worker had been in different physical or emotional condition before the accident
- Workers who smoke, drink, or do not get physical exercise are still entitled to WC benefits for the occupational injuries
  - Example: diabetic patient with a laceration and develops a severe infection

Cumulative Injuries & Illnesses

- Relationship of the work activity to the disability may not be as obvious
- Need to determine the kind of exposure involved (names or types of chemicals, activities involving repetitive motion, etc.)
- The level, frequency, and duration of exposure
- The presenting signs or symptoms that are consistent or inconsistent with the occupational exposure and the disease
- The medical literature, including epidemiological or toxicological studies and case reports that indicate that the disease in question is associated with the worker's exposure or occupation

Causation Analysis

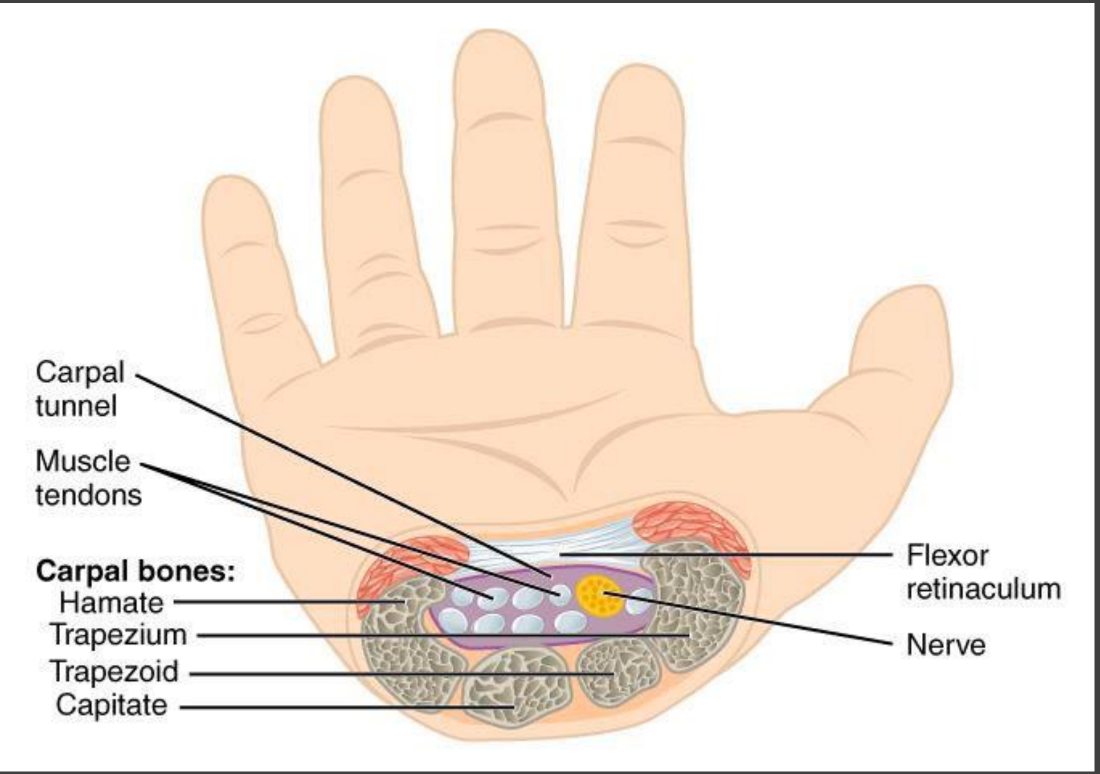
- Mechanism of Injury
- Competing Factors
- Temporality
- Coherence of Association
- Plausibility
- Multiple Contributors
- Reversibility
- Validity of Testimony

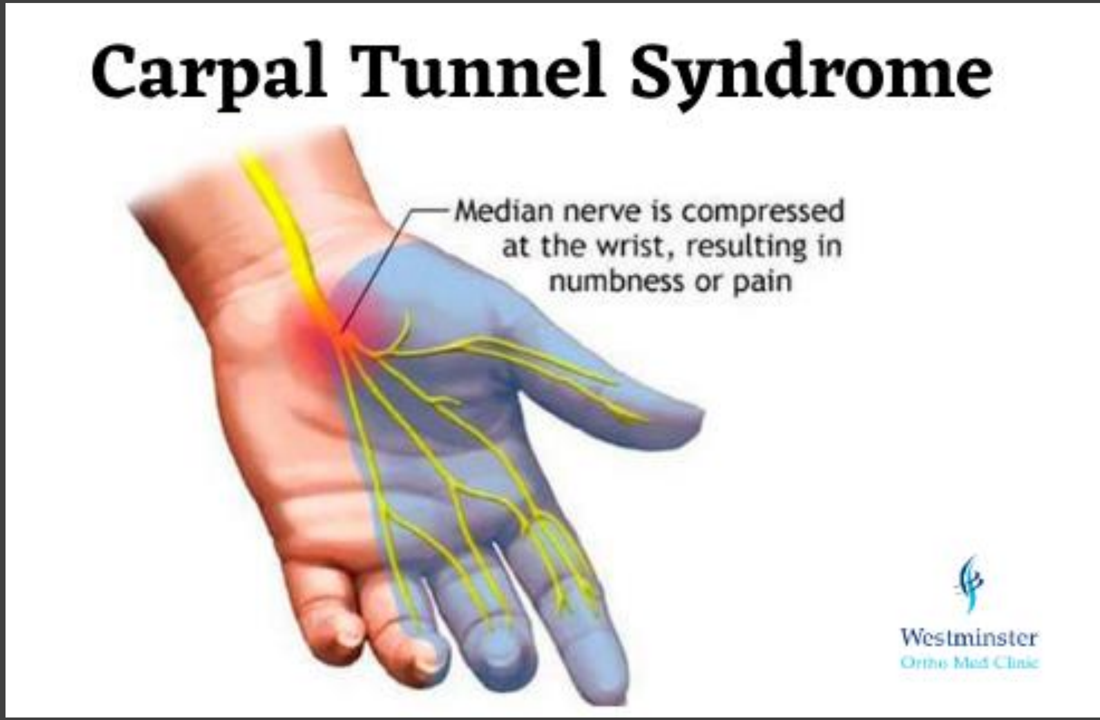
Austin Bradford Hill, "The Environment and Disease: Association or Causation," Proceedings of the Royal Society of Medicine, 58(1965),295-300. By Sir Austin Bradford Hill

Causation Analysis

Shoulder pain while reaching across a desk didn't cause that rotator cuff tear, and bending over to pick up a wallet shouldn't lead to a spine consult…

- Mechanism of injury
  - Plausibility
- Competing factors
  - Multiple contributors
  - Temporality
  - Reversibility
- Coherence of association
  - Validity



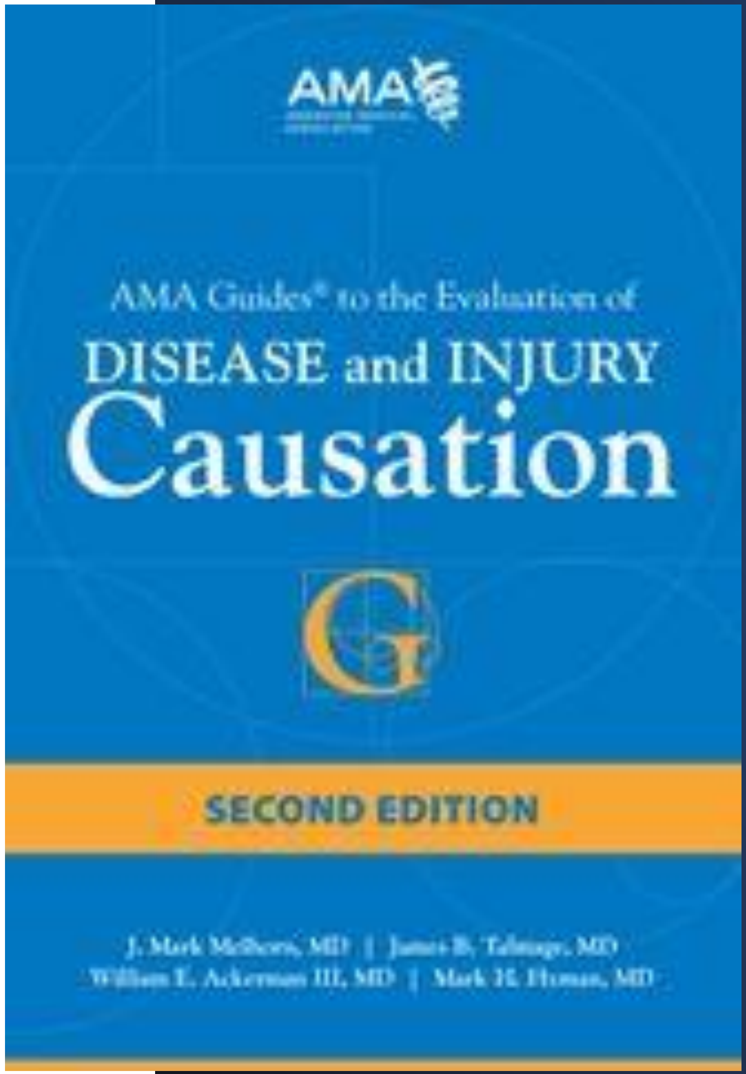


Carpal Tunnel Syndrome

- Entrapment of the median nerve at the wrist is the most common peripheral neuropathy, and one of the most common work injury claims but…

Medical Literature

- Occurs in auto assembly workers, jackhammer operators, and probably in other jobs that require sufficient prolonged or very heavy and repetitive grasping and torqueing
- Vibration alone doesn't cause it



AMA Guides to the Evaluation of Disease and Injury Causation, Second Edition, P. 282-283:

- Report very strong evidence for occupational risk factors from force and repetition however lists jobs with high upper extremity demands such as poultry processing and auto assembly work. There is **insufficient evidence** to support keyboard activities and low risk evidence to support awkward postures. On the other hand, there is **very strong evidence** to support CTS as increasing with BMI and age. Table 9-21 P. 284-292 provides details of risk factors and findings from multiple clinical studies. On P. 293 “An individual who says ‘My job caused my carpal tunnel” may be correct if the job title is chainsaw logger or stone quarry drill operator and job activities require a minimum of 8 hours of chainsaw or stone drill use per day for at least 20 hours a week for an aggregate of – 24 months prior to the onset of symptoms. (Vibration alone had low risk evidence however P. 285).

Meta-analysis on epidemiological studies was undertaken to assess association between carpal tunnel syndrome (CTS) and computer work

- METHODS:
  - Four databases (PubMed, Embase, Web of Science, and Base de Donneesde SantePublique) were searched with cross-references from published reviews. We included recent studies, original epidemiological studies for which the association was assessed with blind reviewing with control group. Relevant associations were extracted, and a metariskwas calculated using the generic variance approach (meta-odds ratio [meta-OR])
- RESULTS:
  - Six studies met the criteria for inclusion. Results are contradictory because of heterogeneous work exposure. The meta-OR for computer use was 1.67 (95% confidence interval [CI], 0.79 to 3.55). The meta-OR for keyboarding was 1.11 (95% CI, 0.62 to 1.98) and for mouse 1.94 (95% CI, 0.90 to 4.21).
- CONCLUSION:
  - It was not possible to show an association between computer use and CTS, although some particular work circumstances may be associated with CTS.

J OccupEnviron Med. 2014 Feb;56(2):204-8. Is carpal tunnel syndrome related to computer exposure at work? A review and meta-analysis.
28 MediouniZ1, de RoquemaurelA, DumontierC, BecourB, GarrabeH, RoquelaureY, DescathaA.

So…Is Carpal Tunnel Syndrome Work-Related?

- “The debate is less about whether medicine is science. It’s about when medicine should be an art and when it is science.”

Presumptions

The law defines specific conditions (hernias, pneumonia, TB, heart disease, and cancer) as work related when they affect certain employees (firefighters, forestry officers, peace officers, and correctional employees)

The medical conditions are presumed to AOE and in the COE

Covers conditions that manifest or develop during the period of active service and following termination of service for up to 5 years

Rebuttable Presumption: assumption that can be contradicted by provided evidence to the contrary

Psychiatric Injuries

- An injured worker must prove that the “actual events of employment” were the “predominant cause” (presumed to be more than 50%) among all the combined causes of the psychiatric injury.
- For psychiatric injuries that result from a violent act or from direct exposure to a significant violent act, the actual events of employment must have been a “substantial cause” of the injury in that they contributed at least 35% of the causation from all sources combined
- Claims filed after notification of termination or layoff are prohibited

Psychiatric Injuries continued

- A psychiatric injury is not compensable unless the employee was employed by the employer for at least 6 months, which need not have been continuous.
- Claims for psychiatric injuries that are substantially (at least 35%) caused by “lawful, non-discriminatory, good faith personnel actions” are prohibited. This prohibition is meant to eliminate claims that were filed by workers who suffered stress resulting from personnel action, such as being passed over for promotion or being transferred to another department.

Physician must consider the following:

- Physicians must take a much more detailed history when evaluating stress claims
- Employee's developmental history
- Personal problems
- Job satisfaction
- Performance reviews
- Reasons for leaving other positions
- Psychiatric history – home, academic, social settings

Case Study

- 57-year-old male who works as a equipment installer for military ships presents with swollen hands for several weeks after pulling cables through tight spaces.

He also has noticed a rash that has been present for several months.


HANES
36

- Hand X-rays


L
PL

Types of Pre-Employment (Pre-Placement) Exams

- Pre-Employment Exams
- Fitness for Duty or Return to Work
- Regulatory OSHA Surveillance – environmental hazards
- Transportation Safety – FAA, DOT, Railroads
- Specific Jobs – Law Enforcement, Fire Fighters, Merchant Marines, Coast Guards
- Immigration Physicals and Travel Medicine

Pre-Placement Exams

This is not “just” a basic physical exam

Evaluating the candidate to be able to do the essential job functions of the job

Job Description vs. Functional Job Analysis

Clinical findings based on history and exam

Understanding Preventable Risk Factors

Protecting both employers and employee

Fitness for Duty & Return to Work Exams

Fitness for Duty: A medical examination of a current employee to determine whether the employee is physically and/or psychologically able to safely perform the job.

Return to Work: A fitness-for-duty examination required when an employee is ready to return to work after taking time off for a serious illness or injury.

Regulatory OSHA Surveillance Examinations

- Know guidelines and interpretations of standards and laws as pertains to the scope of regulatory exams
- Respiratory Protection, Bloodborne Pathogen Exposure, Asbestos and Lead Standards
- Provides recommendations specific to worksite examinations including safety issues

Regulatory OSHA Surveillance Examinations

- Arsenic
- Asbestos
- Benzene
- Beryllium
- Butadiene
- Cadmium
- Chromium
- Diacetyl
- Ethylene Oxide
- Formaldehyde
- Hearing and noise
- Heat and hot environment
- Isocyanate
- Lead
- Mercury
- Methylene Chloride
- Silica
- Vinyl Chloride

OSHA Surveillance

Centers for Disease Control and Prevention
MMWR
Morbidity and Mortality Weekly Report
Weekly / Vol. 68 / No. 38 September 27, 2019

Severe Silicosis in Engineered Stone Fabrication Workers — California, Colorado, Texas, and Washington, 2017–2019

Cecile Rose, MD[1,2*]; Amy Heinzerling, MD[3,4*]; Ketki Patel, MD, PhD[3]; Coralynn Sack, MD[6,7]; Jenna Wolff[1]; Lauren Zell-Baran, MPH[1,8]; David Weissman, MD[9]; Emily Hall, MPH[5]; Robbie Sooriash, MD[5]; Ronda B. McCarthy, MD[10]; Heidi Bojes, PhD[5]; Brian Korotzer, MD[11]; Jennifer Flattery, MPH[3]; Justine Lew Weinberg, MSEHS[3,12]; Joshua Potocko, MD[13]; Kirk D. Jones, MD[14]; Carolyn K. Reeb-Whitaker, MS[15]; Nicholas K. Reul, MD[6,7,16]; Claire R. LaSee, MPH, MSW[15]; Barbara L. Materna, PhD[3]; Ganesh Raghu, MD[6]; Robert Harrison, MD[3]

Silicosis is an incurable occupational lung disease caused by inhaling particles of respirable crystalline silica. These particles trigger inflammation and fibrosis in the lungs, leading to progressive, irreversible, and potentially disabling disease. Silica exposure is also associated with increased risk for lung infection (notably, tuberculosis), lung cancer, emphysema, autoimmune diseases, and kidney disease (*1*). Because quartz, a type of crystalline silica, is commonly found in stone, workers who cut, polish, or grind stone materials can be exposed to silica dust. Recently, silicosis outbreaks have been reported in several countries among workers who cut and finish stone slabs for countertops, a process known as stone fabrication (*2–5*). Most worked with engineered stone, a manufactured, quartz-based composite material that can contain >90% crystalline silica (*6*). This report describes 18 cases of silicosis, including the first two fatalities reported in the United States, among workers in the stone fabrication industry in California, Colorado, Texas, and Washington. Several patients had severe progressive disease, and some had associated autoimmune diseases and latent tuberculosis infection. Cases were identified through independent investigations in each state and confirmed based on computed tomography (CT) scan of the chest or lung biopsy findings. Silica dust exposure reduction and effective regulatory enforcement, along with enhanced workplace medical and public health surveillance, are urgently needed to address the emerging public health threat of silicosis in the stone fabrication industry.

Investigation and Results

California. In January 2019, the California Department of Public Health identified, through review of hospital discharge data for silicosis diagnoses (*International Classification of Diseases, Tenth Revision* [ICD-10] code J62.8), a Hispanic man aged 37 years who was hospitalized in 2017 (CA-1) (Table). He worked at a stone countertop fabrication company during 2004–2013, mainly with engineered stone. His work tasks included polishing slabs and dry-cutting and grinding stone edges. Workplace measurements during a California Division of Occupational Safety and Health inspection in 2009 showed respirable crystalline silica levels up to 22 times higher than the permissible exposure limit (PEL) of 0.1 mg/m^3 in effect in California at that time.† After developing respiratory symptoms in 2012, he had a chest CT scan, which revealed findings of silicosis. Pulmonary function testing showed restrictive defects with reduced diffusion capacity; surgical lung biopsy showed mixed dust pneumoconiosis with polarizable particles

* These authors contributed equally to the report.

† A permissible exposure limit (PEL) is the highest permissible level of exposure for a specific substance for an employee, as established under state or federal occupational safety and health regulations. The PEL cited here is for exposure as an 8-hour time-weighted average, which represents an employee's average airborne exposure to a particular substance during an 8-hour work shift.

INSIDE

Continuing Education examination available at https://www.cdc.gov/mmwr/cme/conted_info.html#weekly

U.S. Department of Health and Human Services
Centers for Disease Control and Prevention

FRA

Federal Railroad Administration

FMCSA

Federal Motor Carrier Safety Administration

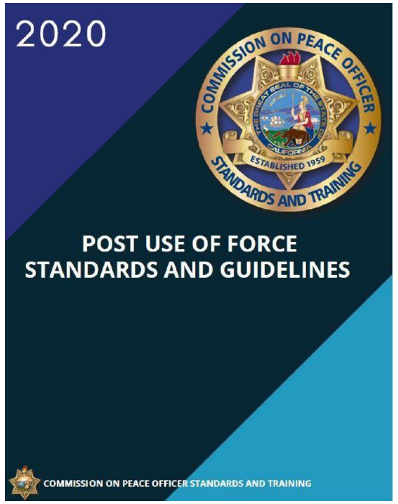
Federal Aviation Administration

Transportation Physicals

Transportation Physicals - OSA

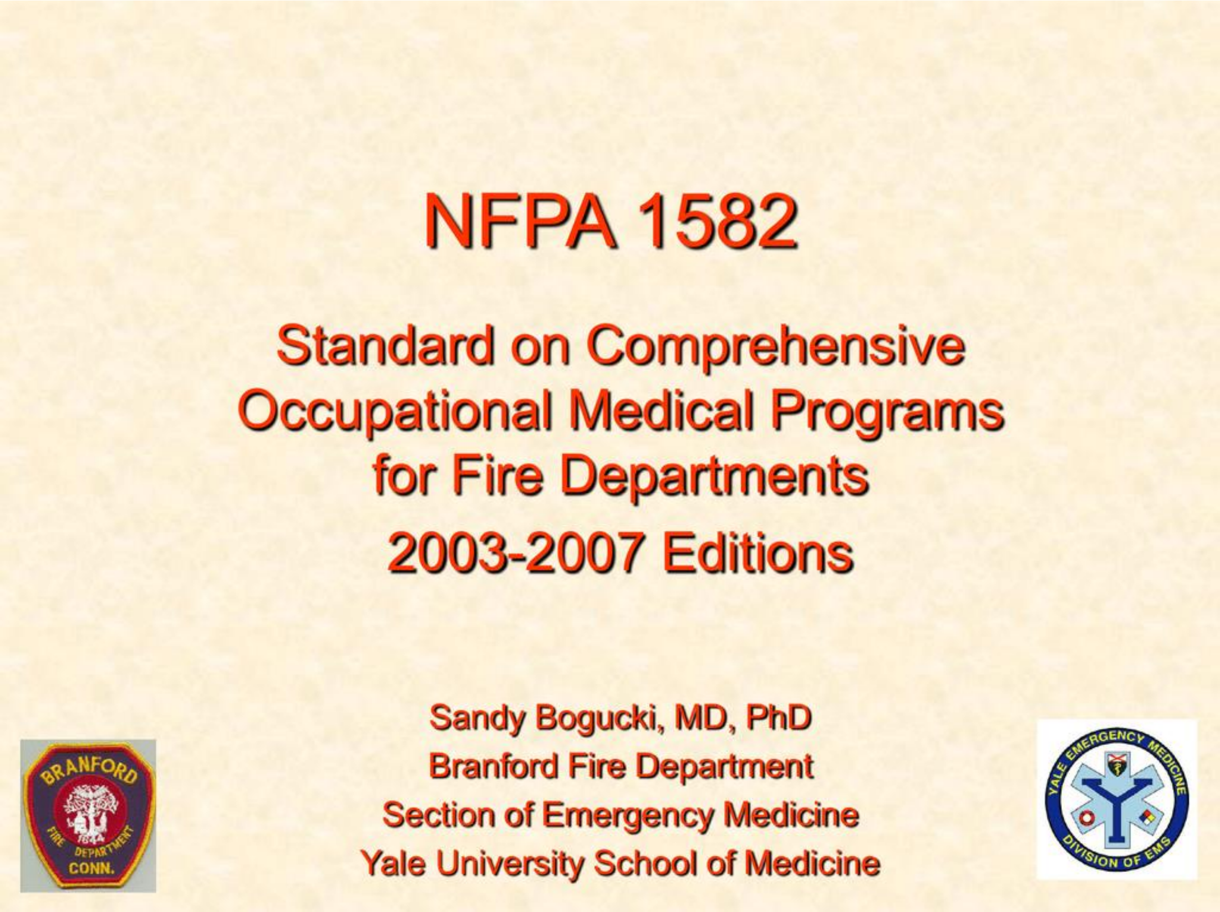
- DOT Standard Section 391.41(b)(5): A driver may operate a commercial motor vehicle if the driver has no established medical history or clinical diagnosis of a respiratory dysfunction likely to interfere with his/her ability to control and drive a commercial motor vehicle safely
- Cost (whose responsibility to burden cost for screening) vs. Public Safety
- Legal issues and Litigation

- Considerations:
- Public Safety
- Long term health effects of the job
- Life and Death situations
- Obesity due sedentary nature of some work



Guidance for the
Medical Evaluation of Law Enforcement Officers
provided by ACOEM

Synopsis of NFPA 1582

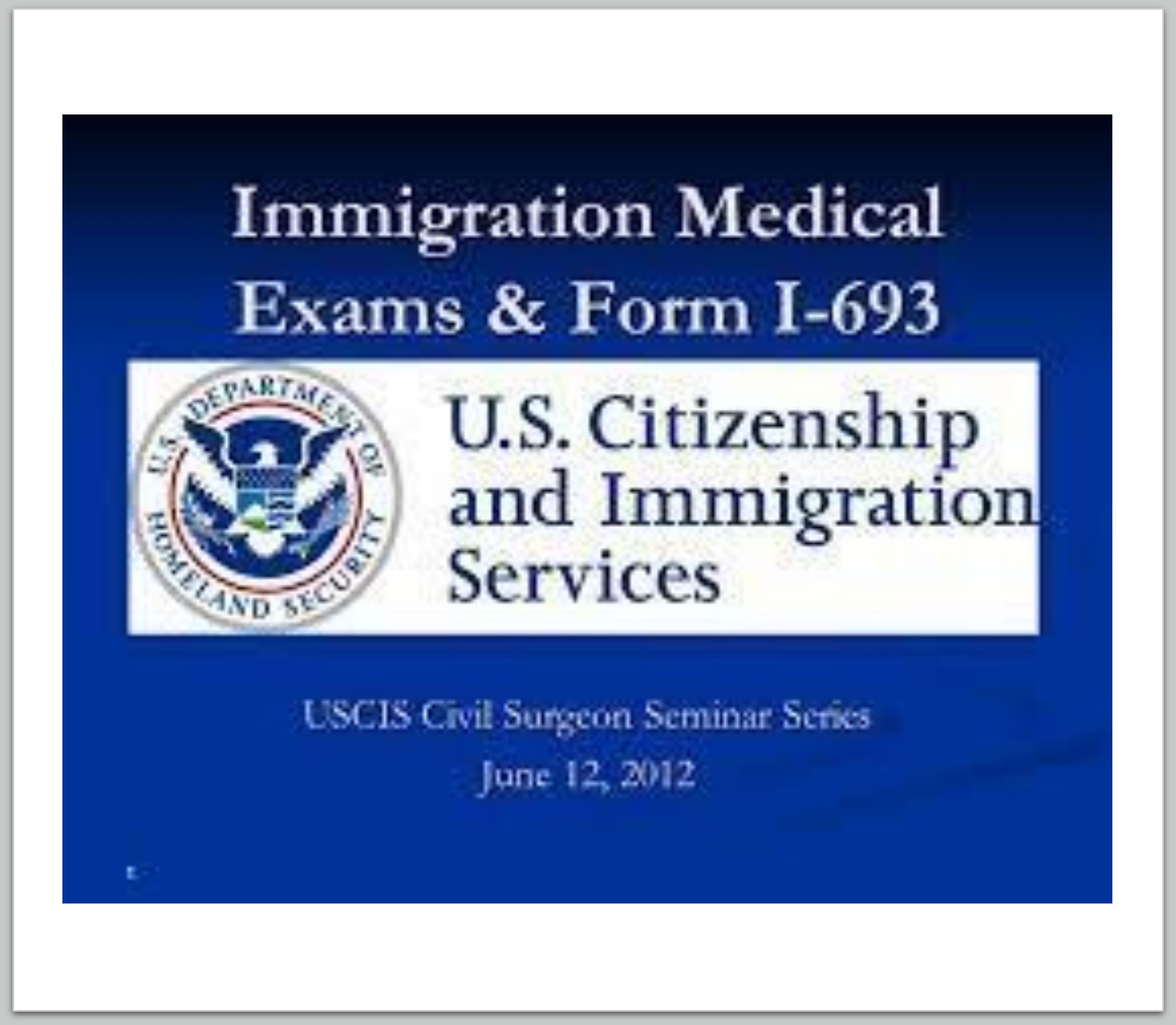


The standard's purpose is to reduce the RISK of fire service occupational morbidity and mortality while improving the safety and efficiency of firefighters and addresses medical issues of both incumbents and candidates

Most common cause of death in firefighters is CANCER

Immigration Physicals

- Considerations:
  - Public Health Issues
  - TB Status
  - Infectious diseases
  - General health
  - Mental health



Travel Medicine

- The goal of a pre-travel medical evaluation is to help travelers protect themselves against
- Common diseases that may be mild but that will disrupt their trip
- Less common diseases that may be serious or even fatal

Thank you for your attention